24:9
Gear Pinion

Equations

Eqs for Power transmitting shafts (English)

$$P = \frac{D^3 N}{80} \quad \text{or} \quad D = \sqrt[3]{\frac{80P}{N}}$$

P = Power in HP
D = Diameter (in)
N = rotational speed (rpm)

Eqs for Power transmitting shafts (SI)

$$P = \frac{D^3 N}{1.72 \times 10^6} \quad \text{or} \quad D = \sqrt[3]{\frac{1.72 \times 10^6 P}{N}}$$

P = Power (kw)
D = Diameter (mm)
N = Rotational speed (rpm)

Eqs for Short Shafts

(English)

$$P = \frac{D^3 N}{38} \quad \text{or} \quad D = \sqrt[3]{\frac{38P}{N}}$$

(SI)

$$P = \frac{D^3 N}{0.83 \times 10^6} \quad \text{or} \quad D = \sqrt[3]{\frac{0.83 \times 10^6 P}{N}}$$

Torsional Strength of Shafting (N-mm or lb-in)

(English)

$$T = \frac{63{,}000\,P}{N}$$

(SI)

$$T = \frac{9.55 \times 10^6 P}{N}$$

$$T \leq \tau_s \cdot Z_p$$

$$\tau_s = \frac{\tau_{in}}{Z_p}$$

Polar section modulus (Z_p)

for ●

$$\frac{\pi D^3}{16}$$